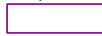
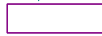


Daisy



File: Seed3-DevKit-Desktop_1.kicad_sch

Peripherals

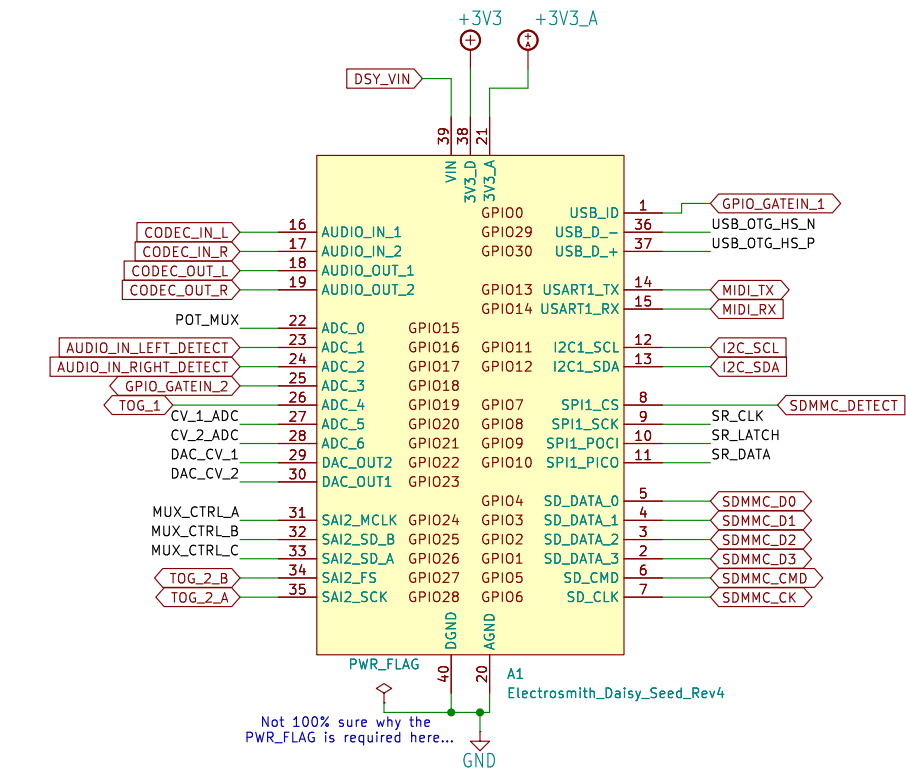
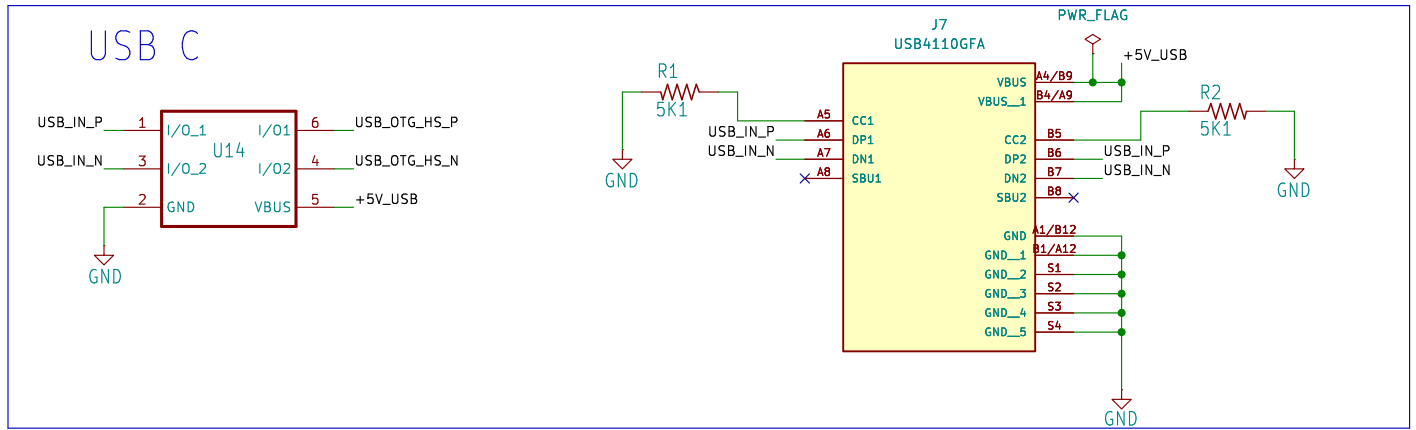
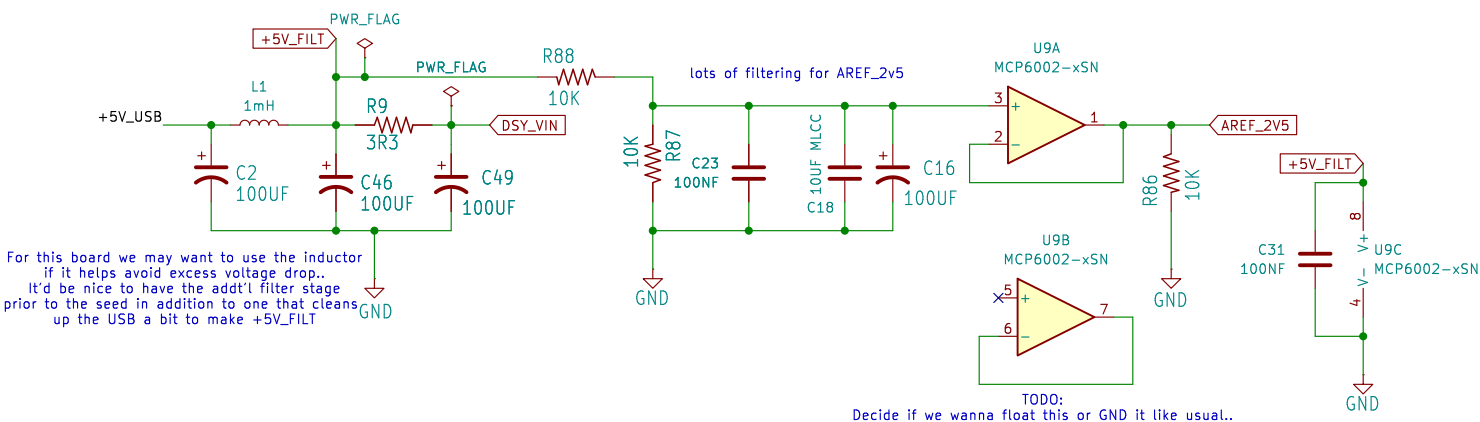


File: Seed3-DevKit-Desktop_2.kicad_sch

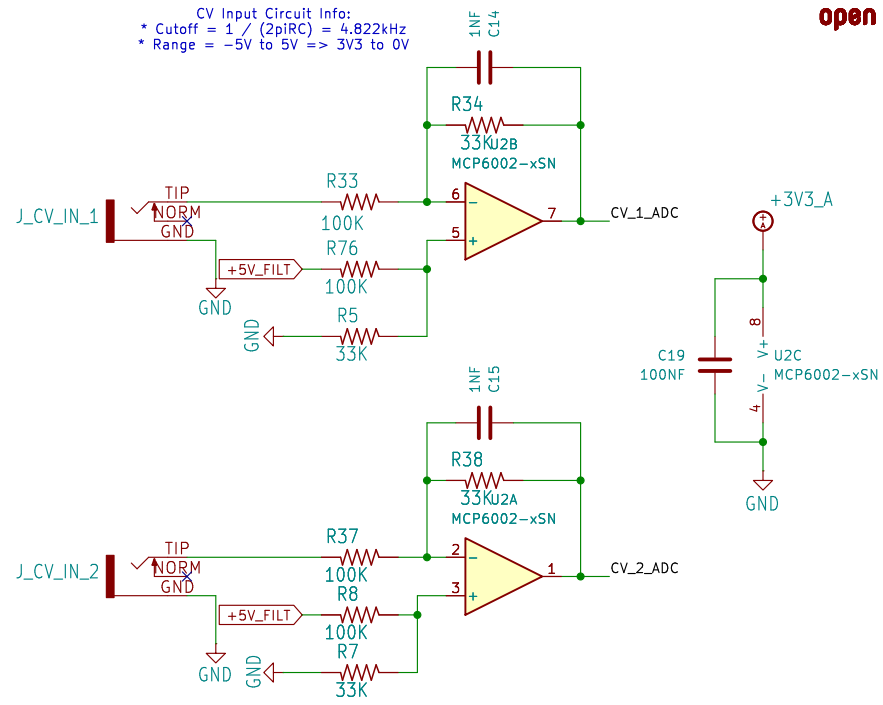
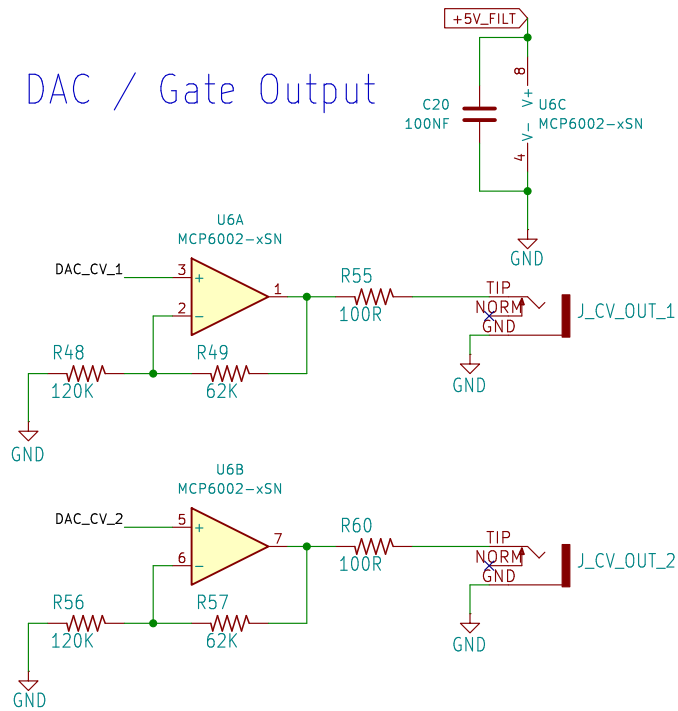
Audio



File: audio.kicad_sch



DAC / Gate Output



[illegible]

LEDs

The diagram illustrates 16 individual LED connections, labeled LED_1 through LED_16. Each LED is connected to a common ground (GND) through a 2K resistor. The connections are as follows:

- LED_1: R43, D1 LED
- LED_2: R35, D2 LED
- LED_3: R36, D3 LED
- LED_4: R39, D4 LED
- LED_5: R40, D5 LED
- LED_6: R41, D6 LED
- LED_7: R42, D7 LED
- LED_8: R44, D8 LED
- LED_9: R45, D9 LED
- LED_10: R46, D10 LED
- LED_11: R47, D11 LED
- LED_12: R58, D12 LED
- LED_13: R69, D13 LED
- LED_14: R70, D14 LED
- LED_15: R71, D15 LED
- LED_16: R72, D16 LED

MIDI

The diagrams illustrate the MIDI circuitry for the device, showing connections for MIDI IN, MIDI OUT, and MIDI THRU.

MIDI IN: The MIDI IN signal is connected to the input of the 74VHC14 inverter (U11). The input is pulled up to +3V3 by resistor R4 (220R). The output of the inverter is connected to the MIDI RX pin. The inverter is decoupled with capacitor C1 (100nF) to ground.

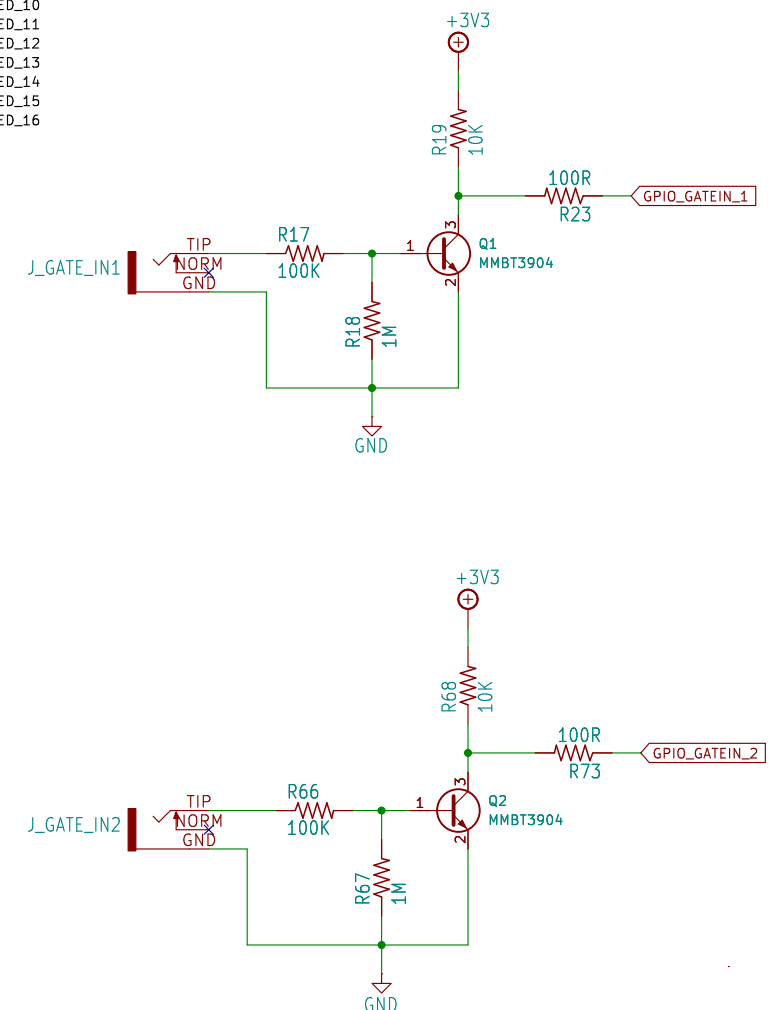
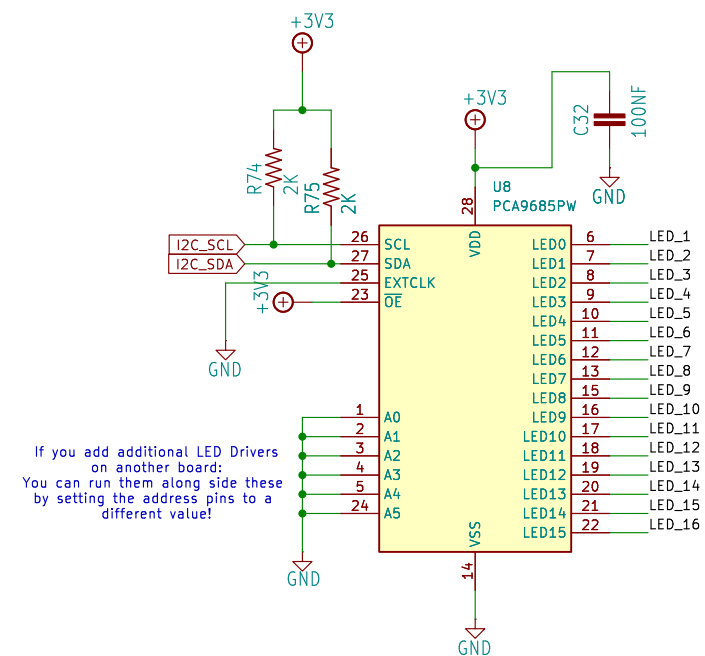
MIDI OUT: The MIDI TX signal is connected to the input of the MIDI OUT connector (J11). The input is pulled up to +3V3 by resistor R6 (10R). The output of the connector is connected to the MIDI OUT pin.

MIDI IN and THRU: The MIDI IN signal is connected to the input of the MIDI IN connector (J10). The input is pulled up to +3V3 by resistor R10 (10R). The output of the connector is connected to the MIDI IN pin. The MIDI RX signal is connected to the input of the MIDI THRU connector (J8). The input is pulled up to +3V3 by resistor R10 (10R). The output of the connector is connected to the MIDI THRU pin.

Power and Ground: The +3V3 power supply is connected to the VCC pin of the inverter and the pull-up resistors. Ground is connected to the GND pin of the inverter and the ground pins of the connectors.

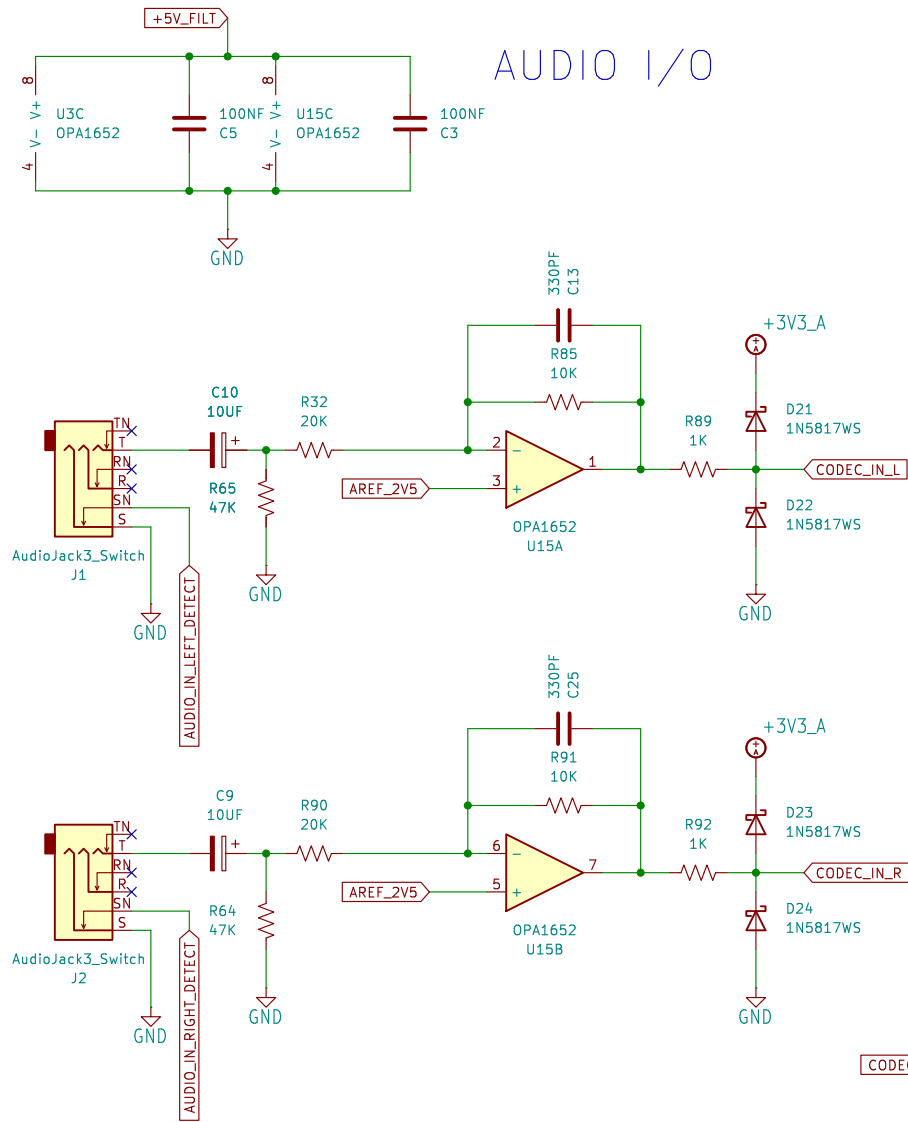
Notes:

- The PWR_FLAG pin is connected to the output of the inverter (U11).
- The MIDI IN and THRU connectors are 5-pin D-sub connectors.
- The MIDI OUT connector is a 3-pin D-sub connector.



The diagram illustrates the wiring for an SD Card interface using a MICRO_SD7 module. The module is connected to a +3V3 supply and a TF-01A card. The module's VDD pin is connected to +3V3, CD1 pin is connected to SDDMMC_DET, and MPB pin is connected to ground. The module's CLK pin is connected to SDDMMC_CK, CMD pin to SDDMMC_CMD, DAT0 pin to SDDMMC_D0, DAT1 pin to SDDMMC_D1, DAT2 pin to SDDMMC_D2, and CD/DAT3 pin to SDDMMC_D3. Pull-up resistors (R50, R51, R52, R53, R54) are connected to +3V3 and the module's DAT0, DAT1, DAT2, and CD/DAT3 pins respectively.

AUDIO I/O

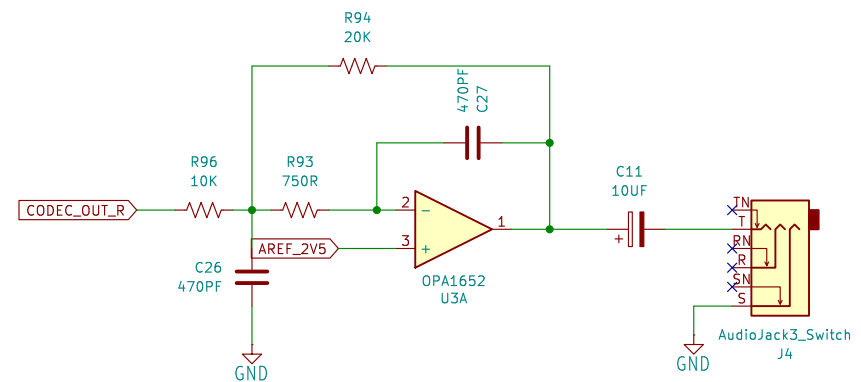


Gain is set to target +4dBu (maximum) obviously this does not make it a pro-line-level device that way.. we'd need to probably have a certain amount of headroom. If, instead we look at this as a -10dBV device, then we have -14dB of headroom with these values.

$G_{in}=0.814$ ($R_{in}=10k$, $R_{fb}=16k$)
 $G_{out}=1.228$ ($R_{in}=10k$, $R_{fb}=24k$)

Alternatively we can go closer to the PCM3060's "2Vrms" reference and just do:
 $G_{in}=0.5$, $G_{out}=2$:shrug:

For now, I've implemented the left channel of audio with less atten/gain.



I updated these to be gain of 2.
 However, I'm not really sure how the math works out for the other two values..
 So, I may just need to kludge those a bit IRL.
 (This can also be reworked to be our normal basic filter)

